

Measuring the practice of shared-decision making (OPTION12): An Investigation into Open-sourced Smaller LLMs (OS-sLLMs) for Better Privacy and Sustainability

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Accepted Abstract

Background: Shared decision-making (SDM) is important in clinical consultations in which patients discuss and decide on treatment options with clinicians. This SDM process is often coded with the OPTION12 instrument, an observer-based tool, consisting of 12 items, rated on a 5-point Likert scale. This coding is conducted by human coders. Coding is time-intensive and frequently accompanied by disagreement between coders. We explore the capability of open-source privacy-preserving smaller LLMs (OS-sLLMs) to perform the coding task, potentially automating the process, with humans in the loop. **Methods:** 26 transcripts of Dutch melanoma patients consultations with clinicians were double-coded by two coders. Two human coders resolved their disagreements after independent coding. To evaluate OPTION12 coding using OS-sLLMs, the consultation data was divided into development and testing sets. We designed the complete investigation framework (see Figure 1). It includes 1) a pilot study of the development set (11 interviews) for prompt-refinement and sLLM selection. 2) deployment of the fine-tuned prompts and best performing sLLM (judge-sLLM) augmented with few-shot examples from the development set on the testing set (15 interviews) and asking the judge-sLLM to resolve the disagreement on other OS-sLLMs' scores. The rationale for judge-llm: it behaves like a third-party human to resolve the disagreements of annotators. Instead of using humans, a judge-LLM is consulted to resolve disagreement. This is designed to mimic how human annotators resolve disagreements, by discussing until consensus is achieved. We plan to investigate alternative model-merging methods for the future. First, for prompt re-

finement, we use chain-of-thoughts (CoTs), LLM-assisted prompting, human-in-the-loop with sample output (few-shots) feedback. For OS-sLLMs, we use both 1) general domain models Llama, Gemma, and Mistral7b, and 2) medical domain models Meditron and Medllama. Second, for judge-sLLM selection, on the system-level, we measure the overall correlation of each sLLM and human coding using Spearman and Pearson correlation scores. On the segment-level, we also look into the most agreed-upon and disagreed-upon items. We will perform both qualitative and quantitative analysis by discussing the evaluation scores and categorise the OS-sLLMs' behaviours with examples. Finally, we deploy the OS-sLLMs on the testing set and use the judge-sLLM to resolve disagreements.

Preliminary results: Five OS-sLLMs show the following findings:

- Three general domain OS-sLLMs perform better than the two medical domain ones, which both generate hallucinations and do not follow prompts precisely, indicating further developments is needed for medical OS-sLLMs.
- Mistral7b outperformed the other two Gemma3:12b and Llama3.1:8b by 4 consensus with human coding, vs 3 items.
- The overall correlation with human coding on these 12 items is (0.83, 0.80, 0.64) using Pearson correlation, and (0.81, 0.78, 0.61) using Spearman rank correlation from the three models (gemma3:12b, llama3.1:8b, mistral7b).
- For the items for which OS-sLLMs agree with human coders, OS-sLLMs can generate the same sentences as humans in

some cases, but in other cases, they generate better quotes than humans.

Impacts: This study reports the first research findings using OS-sLLMs on scoring SDM with the OPION12 in Dutch melanoma patients' consultation transcripts. The performance of such OS-sLLMs is promising and valuable to our task on measuring patient involvement. They have agreements with human coders on certain items and we are looking into the disagreement to see if we can have different inputs from OS-sLLMs, or if we can fine-tune OS-sLLMs to achieve human level performances on the rest of the items. In the long-term view, we expect fine-tuned OS-sLLMs can achieve human level performance and be deployed for coding tasks with humans in the loop for verification and quality control. ¹

Extended work from Conf Abstract

1 Introduction

Shared Decision Making (SDM) is very important for healthcare since it encourages patients and their family members to get involved in the decision-making process of their treatments (Stiggelbout et al., 2015). There are regulations for SDM guidelines, such as OPTION12; however, there is the coding burden from human annotators and such manual work.

There are articles on using LLMs for medical decision making collaborations, but rarely on predicting the level of SDM in practice, with the only one we found on breast cancer only (Selvaraj et al., 2025). It is on a different coding scheme (OPTION5), and it uses commercial GPT models (GPT-4o, Gemini-1.50pro) that are very large, costly, and have potential privacy/ethical concerns regarding patient/medical data usage.

To promote sustainable development, smaller-sized LLMs need to be tested for such a task (e.g. instead of Llama 405b, Llama 70b); to protect privacy, open-sourced, local LLMs need to be researched and deployed. There is a research Gap: how to measure OPTION12 SDM performance using open-source models for better privacy. In our case, we have Dutch melanoma patient data.

OUR research objective: How open-sourced LLMs perform in predicting the SDM implementation levels in the clinic and if their performance can be improved with fine-tuning using human-annotated data; furthermore, if LLMs can be integrated into this task to save manual annotation cost.

Research Questions: **RQ-1a:** if state of the art open-sourced smaller LLMs (OS-sLLMs) can perform human-like annotation regarding the levels of SDM implementations in clinical practice? **RQ-1b:** 1) if LLMs perform differently with human annotators, do they give a different point of view, or just nonsense? 2) if they score the same as humans, do they pick the same quotes? **RQ-1c:** which of the 12 items in OPTION12 do LLMs tend to perform better/worse? what are the overall correlation levels with human coding? **RQ-1d:** do general domain LLM perform differently vs medical domain LLMs?

¹extended work based on accepted Abstract at ISDM2026 <https://sites.dartmouth.edu/isdm2026>

RQ-2: can fine-tuning LLMs (or example-based learning) using human annotated data help them on performing this specific task on SDM implementation levels from Scheme-OPTION12?

RQ-3: can we use LLMs to conduct this annotation task if it can perform fairly, e.g. LLMs + human in the loop, so as to save cost?

Highlighted contributions of this work are:

- First investigation of open-source small LLMs for OPTION12 coding.
- Comparison of: general-domain OS-sLLMs and medical-domain OS-sLLMs for SDM assessment.
- Discovery that: medical models hallucinate more; general models perform better; despite domain specialization (a surprising result).
- Identification of systematic SDM-coding error categories (taxonomy): temporality failures, role confusion, evidence hallucination, score-evidence mismatch, option misunderstanding.
- Proposal of the Judge-LLM consensus framework.

2 Related Work

We describe related and background work in SDM and OPTION scale, regional or domain-specific SDM, and LLM investigation for SDM.

2.1 SDM and the OPTION scale

The early works on developing the OPTION scheme for SDM include (Elwyn et al., 2005), where the authors designed the scales to measure patient involvement in the SDM procedure (Elwyn et al., 2003).

Later, there are inspired academic discussions and adaptations of such designs into different sub-fields in healthcare, for instance, SDM in Psychometric instrument (Simon et al., 2007) and psychometric study of OPTION5 (Barr et al., 2015); dyatic OPTION for perceptions of SDM (Melbourne et al., 2010); comparing OPTION and informed SDM instruments (Weiss and Peters, 2008); discussing methodological issues of the OPTION scale (Nicolai et al., 2012); OPTION - convergent validity

(Scholl et al., 2015); OPTION5 vs OPTION12 (Stubenrouch et al., 2016) comparison. Experimental results from earlier works show that the OPTION5 scores can be consistently higher than the OPTION12 ones. We think this can be caused by the fact that OPTION12 includes more aspects to measure. In our current research project, we use melanoma patients' consultation data on the OPTION12 scales. More SDM overview works can be found at (Scholl et al., 2011; Couët et al., 2015; Gärtner et al., 2018; Ahmad et al., 2020).

2.2 Region and Domain-Specific SDMs

Another line of work related to ours is the regional or domain-specific SDMs, since we are focusing on Dutch Melanoma patients. Regarding regional SDM, such works include the study of the reliability of the OPTION score in Italy (Goss et al., 2007), SDM in Canada (Elwyn et al., 2013). Focusing on specific patients, there are SDM in outpatient chronic care (Norful et al., 2020), SDM in cardiovascular care (Sepucha and Scholl, 2014), as well as OPTION score for resident patient consultations in family medicine (Pellerin et al., 2011).

2.3 LLMs on SDM

LLMs have been suggested by researchers to assist the SDM process by offering lay language adaptation of medical evaluations and evidence as a collaborator role with the clinician (Elwyn et al., 2024).

(Selvaraj et al., 2025) present the first empirical study investigating whether large language models (LLMs) can automate the observer-based assessment of shared decision making (SDM) using the Observer OPTION-5 instrument. The study evaluates commercial LLMs from the GPT, Gemini, and LLaMA families on 287 anonymized transcripts of clinician-patient consultations involving women considering surgery for early-stage breast cancer. Human-rated OPTION-5 scores from a randomized controlled trial were used as the reference standard. The authors tested multiple prompting strategies and assessed agreement between LLM-generated and human-generated scores using correlation analysis.

The study found that GPT-4o and Gemini-1.5-Pro achieved moderate correlations with

human ratings (Pearson correlation approximately ($r \approx 0.6$)), corresponding to roughly 75–80% of the agreement observed between human raters themselves ($r = 0.77$). Performance improved when prompts included detailed item descriptions and examples, suggesting that prompt engineering substantially affects LLM performance on SDM coding tasks. Additionally, the models were able to distinguish between consultations with high versus low SDM implementation levels, indicating potential utility for automated quality assessment.

This work differs from our work in several ways. First, they focus on the Observer OPTION-5 instrument, whereas our study investigates the more fine-grained OPTION-12 coding scheme. Second, their work uses large commercial models (GPT-4o and Gemini-1.5-Pro), while our study evaluates privacy-preserving, open-sourced, smaller LLMs (OS-sLLMs) that can be deployed locally. Third, their dataset concerns breast cancer surgery consultations in English, whereas our work focuses on Dutch melanoma consultations. Finally, while they primarily assess model validity through correlations with human ratings, our study additionally investigates model disagreement patterns, qualitative error categories, and the potential role of judge-LLMs in resolving scoring disagreements.

3 Methodology

The methodology framework is presented in Figure 1, where the upper half is the developmental phase and the lower half is the deployment phase using tuned LLMs. From left to right, we start with transcripts from melanoma patient where a decision took place. These transcripts have been coded by two researchers (CH/TW). All transcripts were individually coded, consensus meetings were held to discuss potential disagreements until consensus was achieved. The data includes three roles DR, CG, and PT. We split the coded data into development (pilot study) and testing sets. For the **development** phase, transcription pre-processing was carried out for NLP purpose. For model configuration, we explored parameter optimization, context-length and temperature adjustment, prompting with Chain-of-Thoughts (CoTs). We select a list of

open-sourced smaller-sized LLMs (OS-sLLMs) covering both general and medical models. Using the selected OS-sLLMs and our model configuration, we produce the automatic system outputs (i.e., annotations and justifications) of SDM scores from each file in the pilot study. We compare the automatic annotations to the human annotations both quantitatively using correlation scores and qualitatively looking into errors or disagreements. At the end of the pilot study phase, we 1) conduct the categorization of LLM-related issues on this task; 2) select the best performing OS-sLLMs as a judge (Judge-sLLM) for the next phase.

In the **testing** phase, as in the lower part of the methodology figure (Figure 1) from both left and right to the middle, we first carry out the same pre-processing for the documents in the test set as the same procedure of the pilot study. Then, we deploy the parameter-optimized OS-sLLMs w/o judge on the test set through few-shot (example-based) and instruction-based fine-tuning to get the automatic annotations. Thirdly, we use Judge-sLLM to resolve the consensus from the outputs produced by other OS-sLLMs to mimic how human annotators would carry out this work. Lastly, the final output from consensus-ed LLMs will be compared with human annotations and evaluated on the test set.

4 Experiments on Development Set

4.1 Data and Models

We have 26 transcriptions of Shared Decision Making consultations of **melanoma** patients, which are **double-coded** by two researchers in the healthcare sector, and agreed on consensus with the OPTION 12 coding scheme (Observing patients' involvement in decision making). They have numerical values (0-to-4) for a list of 12 categories. We applied the Refined Manual Scheme (RMS), a task-oriented scheme, revised from the existing validated coding scheme (on the public website).² The coders carried out coding independently and resolved the initially disagreed-upon coded values by discussion. The final gold reference coded values are the human-agreed ones. We

²<https://each.international/reachresources/option-12/>

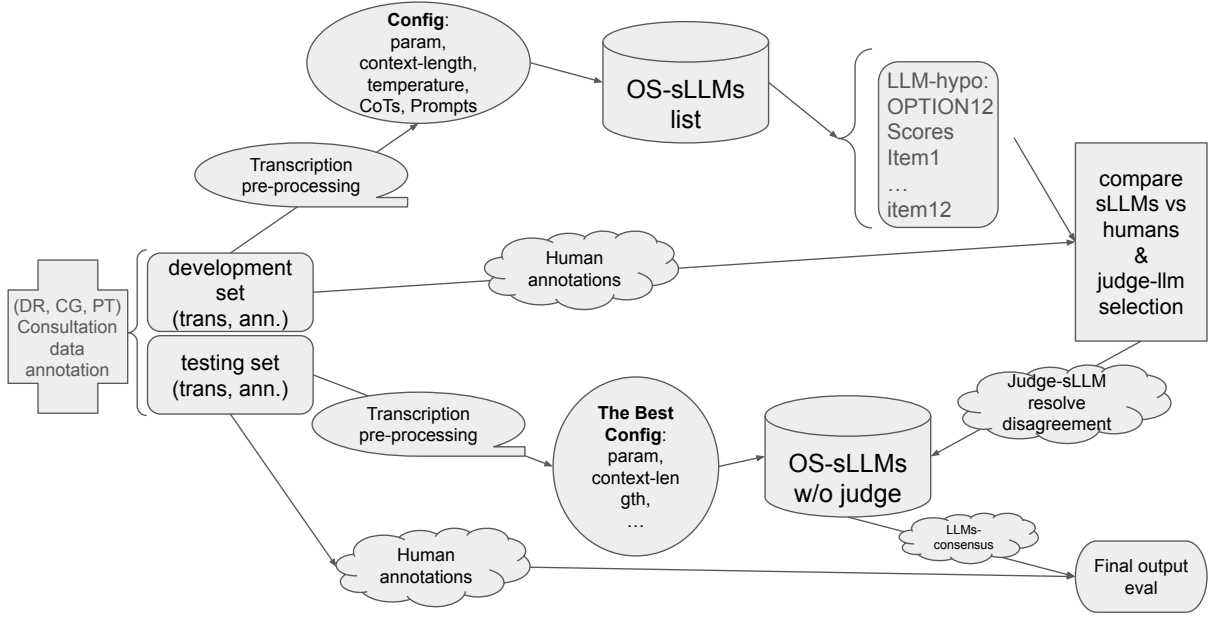


Figure 1: An overview of LLM4SDM framework: upper layer for development phase and lower layer for model testing.

split this data set into development/validation and test sets (11, 15).

To investigate both general and medical domain OS-LLMs, we used the following comparable-sized lightweight models: 1) general domain pre-trained Gemma3:12b, Llama3.1:8b, and Mistral7b, and 2) medical domain finetuned models MedLlama2:7b and Medtron7b.

4.2 Evaluation Metrics

We define several metrics to evaluate the agreement between LLM-generated annotations and human annotations under the OPTION12 scheme.

Total Consensus. The total number of agreements between a model and human annotators across all files and items is defined as:

$$\text{TotalConsensus} = \sum_{f=1}^N \sum_{i=1}^{12} C_{f,i} \quad (1)$$

Mean Consensus per File. The average number of agreed items per file is computed as:

$$\text{MeanConsensus} = \frac{1}{N} \sum_{f=1}^N \sum_{i=1}^{12} C_{f,i} \quad (2)$$

Normalized Mean Consensus. To obtain a percentage-based metric, we normalize the mean consensus by the total number of items:

$$\text{MeanConsensus}_{\text{norm}} = \frac{1}{12N} \sum_{f=1}^N \sum_{i=1}^{12} C_{f,i} \times 100 \quad (3)$$

Per-item Agreement Ratio. For each model m and item i , we compute the agreement ratio across all files:

$$\text{Ratio}_{m,i} = \frac{C_{m,i}}{N}, \quad C_{m,i} = \sum_{f=1}^N \mathbb{I}(\hat{y}_{m,f,i} = y_{f,i}) \quad (4)$$

where $C_{f,i} \in \{0,1\}$ indicates whether the model's prediction matches the human annotation for file f and item i , $C_{m,i}$ is the total number of correct predictions made by model m for item i , N is the number of files (patients), and $\mathbb{I}(\cdot)$ is an indicator function that equals 1 when the condition holds and 0 otherwise.

In practice, we report $\text{Ratio}_{m,i}$ as percentages.

We normalize total agreement counts per item by the total number of model-file combinations ($M \times N$) to obtain percentage agreement rates.


```
{
  'item_number': 10,
  'item_description': 'Clinical assessment',
  'score': 3,
  'evidence': 'The patient was assessed by a dermatologist '
    'and the diagnosis of melanoma was confirmed. '
    'The Breslow thickness was 4 mm.',
  'justification': 'This patient was seen in our clinic for '
    'the first time with a suspicious skin '
    'lesion on his left arm. A biopsy was '
    'performed, and the diagnosis of '
    'melanoma was confirmed by '
    'histopathology. The Breslow thickness '
    'was 4 mm.'
}
```

Figure 2: Meditron Model Hallucination Example - repeated evidence and justification across predicted OPTIOIN12 items using the same score 3.

```
{
  'item_number': 10,
  'item_description': 'Hoeveel pijn voel je?',
  'score': 0,
  'evidence': 'verbal',
  'justification': 'Asking the patient how much pain he '
    'feels is a standard question to ask in '
    'order to assess his general condition. '
    'A score of 1 indicates that the patient '
    'does not feel any pain.'
}
```

Figure 3: Meditron Model Hallucination Example - giving some score 0 for all OPTION12 items but using similar text to justify score 1 instead.

$$\text{TotalCorrectRate}_i = \frac{100}{MN} \sum_{m=1}^M \sum_{f=1}^N \mathbb{1}(\hat{y}_{m,f,i} = y_{f,i}) \quad (5)$$

4.3 Pilot Study Outcomes

To answer our **RQ-1d**: Do medical and general domain LLMs perform differently? Yes, the medical LLMs have more hallucinations.

4.3.1 Hallucinations of Medical LLMs

The two medical OS-sLLMs both generated hallucinated outputs that we have to give up on these tested models: MedLlama and Meditron.

Example of Meditron7b model hallucination behavior on output is show via patient file “130002” in Figure 2, where in the first run, the model generated score 3 for its outputs only covering partial OPTION12 items (Item 10, 11, and 12) using the same repeated evidence and justification across these items. However, in the 2nd run, after we fix its missing items in the output by more strict instruction, it generated score 0 for all OPTION12 items (Item 1 to 12) but using similar evidence and justification phases to justify score 1, as in Figure 3.

MedLlama also produced hallucination in the initial deployment when it generated score 1 for all OPTION12 items and even gave a mean score 10 very wrongly by averaging individual

```
{
  'item_number': 12,
  'item_description': 'The patient has a history of '
    'exposure to environmental toxins',
  'score': 1,
  'evidence': 'Environmental toxin exposure diagnosis',
  'justification': 'A history of exposure to environmental '
    'toxins may increase the patient's risk '
    'for developing cancer.'
},
{
  'summary': {
    'mean_score': 10.0,
    'overall_quality': 'Good',
    'strengths': ['History of cancer diagnosis'],
    'weaknesses': ['Moedervlek on his back',
      'Family history of cancer',
      'History of smoking',
      'Excessive alcohol consumption',
      'Obesity',
      'Lack of physical activity',
      'Poor diet',
      'Chronic stress',
      'Poor sleep quality',
      'Environmental toxin exposure']
  },
  'additional_comments': None
}
```

Figure 4: MedLlama Model Hallucination Example - giving score 1 for all OPTION12 items but mean score 10.

scores, as in Figure 4 from example patient file “110019”. However, it did give different justifications and evidences across OPTION12 items.

4.3.2 Metric scores from general LLMs

Out of 11 development files, 3 files did not work out due to formatting, and 1 did not receive human code because it was not suitable for SDM. Adjustment: So, in total, we have 7 patient files (representing 7 clinician-patient consultations) successfully run through the development phase, as in Table 1. The total number of model-file combinations is $M \times N = 3 \times 7 = 21$.

Table 1: Consultation selection for the development phase.

Step	N
Total consultation transcripts	26
Development set	11
Excluded: preprocessing failures	3
Excluded: unsuitable for SDM coding	1
Development consultations included in analysis	7
Testing set (held out)	15

From total agreement counts and normalized agreement rates per OPTION12 item across all (3) models, Table 3, we can see that OPTION Item 7 has the highest agreement (42.9%, 9 counts), while OPTION Items 1 & 8 are the hardest (0%) for all AI models.

The result from each model is shown in Table 2. From Table 2, it shows that Gemma3:12b has the largest number count of consensus (20) with human annotation over 7 files/patients consultations, followed by Llama3.1:8b (17)

OPTION12 item / summary	Gemma3:12b (%)	Llama3.1:8b (%)	Mistral7b (%)
1	0	0	0
2	42.9	42.9	28.6
3	42.9	28.6	14.3
4	28.6	57.1	28.6
5	14.3	28.6	28.6
6	42.9	14.3	0
7	57.1	0	71.4
8	0	0	0
9	14.3	0	0
10	42.9	0	14.3
11	0	14.3	14.3
12	0	57.1	14.3
Total consensus count (over all items across 7 files)	20	17	15
Mean consensus per file (out of 12)	2.86	2.43	2.14
Mean consensus per file (normalized, %)	23.83	20.25	17.83

Table 2: Model-wise agreement per OPTION12 item between LLMs and human annotators across 7 files, with overall summary statistics. Bold indicates the best-performing model(s) for each row.

OPTION12 Item	Total Correct Count	Agreement (%)
1	0	0.0
2	8	38.1
3	6	28.6
4	8	38.1
5	5	23.8
6	4	19.0
7	9	42.9
8	0	0.0
9	1	4.8
10	4	19.0
11	2	9.5
12	5	23.8

Table 3: Total agreement counts and normalized agreement rates per OPTION12 item across all models (3) and files (7), i.e., out of 21 possible agreements per item.

and Mistral7b (15). The mean consensus count per consultation (i.e., approximately how many items) is 2.86, 2.43, and 2.14 for these three models, which means there are, on average, between 2 and 3 items that LLMs agree with human annotators exactly, out of 12 items. The normalized percentage values are 23.83%, 20.25%, and 17.83% for the three LLMs, which are very low, i.e., less than 25% of chance they can be right individually.

From the individual model performance table (2), we can also see that: 1) The highest agreement on the OPTION item 7 is due to higher performances from Gemma3:12b (57.1%) and Mistral7b (71.4%), but Llama3.1:8b has 0% correct on this item. 2) Three models perform very differently across 12 items, each model having 0% scores on some items but not always the same items. 3) OPTION items 2 to 5 never have 0% scores. These findings further verified our methodology design on how to use

model augmentation (e.g., multiple models) to achieve better performance across all OPTION items, using a collective effort of LLMs.

In summary, Table 2 and 3 answered our **RQs** 1a): OS-sLLMs are far from human performances on overall OPTION 12 items except for specific items, e.g., Item-7 (Mistral7b achieving a promising score 71.4%); 1c) OPTION items 1 and 7 are extremely hard for OS-sLLMs, and other items have different difficulty levels across models. 1d) Medical domain OS-sLLMs totally failed, in comparison to general domain ones.

Some further findings from these two tables on OPTION12 items' difficulty level are: 1) Easy: Item 7 with 42.9%; 2) Medium: Items 2, 4, with 38.1%; 3) Hard: Items 1, 8, with 0%. The variation in performance across OPTION12 items suggests that different cognitive and discourse-analysis skills are required for successful coding. For example, Item 1 ("The clinician draws attention to an identified problem as one that requires a decision-making process") requires the model to perform temporal framing and discourse-structure reasoning. The model must identify not only whether a decision is discussed, but also whether the clinician explicitly frames the situation as requiring a decision at the appropriate stage of the consultation. Several models recognized later treatment discussions but failed to distinguish them from the initial decision-framing process.

In contrast, Item 8 ("The clinician checks that the patient has understood the information")

Model	Pearson (r)	Spearman (ρ)
Gemma3:12b	0.513	0.587
Llama3.1:8b	0.180	0.045
Mistral7b	0.180	0.101

Table 4: Correlation between LLM scores and human annotations across OPTION12 items over 7 patient files.

requires interactional reasoning and clinician-verification recognition. Models must correctly identify conversational acts in which the clinician actively assesses patient understanding, distinguish them from general information provision, and avoid confusing statements made by patients or caregivers with clinician verification behaviours. The low performance on this item suggests that current OS-sLLMs struggle with accurately interpreting conversational roles and communicative intentions within clinical dialogues.

In addition to the measurements based on the counting of exact matching score points between LLMs and human, we also explore the sequential correlation between LLM scores and human scores over OPTION12 items. To do so, we further evaluate the alignment between LLM-generated scores and human annotations using Pearson (r) and Spearman (ρ) correlation coefficients across OPTION12 items. As shown in Table 4, Gemma3:12b achieves the highest correlation with human annotations ($r = 0.513$, $\rho = 0.587$), outperforming Llama3.1:8b and Mistral7b. This also shows the moderate correlation level between LLMs and humans, which can be promising for future development on this direction.

Correlation measures whether models preserve relative scoring trends across OPTION12 items, whereas exact agreement requires predicting the precise 0–4 score assigned by human annotators. Therefore, moderate correlations alongside low exact agreement suggest that the models capture broad SDM patterns but remain insufficiently reliable for fully automated coding.

5 Error Analysis from Pilot Study

5.1 Data Preprocessing

From the development set, we also find a data noise issue from human evaluation. For in-

stance, the Caregivers can be CG but also CG1 or CG2 when there are multiple caregivers in the consultation. The initial preprocessing of our method did not take this into account, so the text following CG1 was not extracted for LLM use in the prompting stage. Our thought is that for OPTION12 items for SDM measurement, it concerns more what doctors and patients say, instead of caregivers. For such files, we plan to **investigate whether CG input/text has a significant impact on the SDM**. So, we will relocate these files (with corrected preprocessing and extraction) to the testing phase and remove them from example-based training (to prevent the model from seeing them before testing) to see whether the SDM scores differ substantially. There are 2 such files with CG and CG1, ID 110022/23, indicating two caregivers in the consultations.

There are also three files with different timing formats, e.g., having the timing rounded with ‘[]’ while other files do not. Such files were not correctly preprocessed, so the LLMs had no output. We relocated 3 of such files from development to the test phase.

5.2 LLM Qualitative Error Categorization

In many cases, LLMs can be correct; however, we list the situations when they go wrong with categorization, as in Table 5. We hope this can help future researchers tackle some of the challenges. This list of categories partially reflects our **RQs: 1a**: whether OS-LLMs perform well, and **1b**: whether OS-LLMs give different views.

The categorized LLM-related issues with examples are 1) inability on temporality-management meaning that it matters for the situation on if the information is given/provided in the beginning, middle, or end of the consultation; 2) mis-interpreting the item/option, e.g. understanding *what it is* instead of *a decision is needed* or *what are the options*; 3) giving right scores as human annotators but wrong or inaccurate evidence or justification; 4) justification or evidence are right but scores are different from human annotators; 5) missing justification or evidence; 6) mis-understanding/swapping role-play; and 7) made-up/hallucinated evidence.

5.3 Medical vs General LLMs

Why do medical models fail? This is an interesting finding, because MedLlama and Meditron perform worse than Gemma, Llama, and Mistral. One possible explanation is that: OPTION12 coding is not primarily medical knowledge. It is: discourse analysis, conversational reasoning, and interaction assessment. General LLMs may be better trained on these abilities.

6 Conclusions and Future Work

This study presents the first investigation of open-source smaller language models (OS-sLLMs) for automated assessment of shared decision making using the OPTION12 framework in Dutch melanoma consultations. Using a development set of human-annotated consultations, we evaluated three general-domain models (Gemma3:12b, Llama3.1:8b, and Mistral7b) and two medical-domain models (MedLlama2:7b and Meditron7b). Our findings show that general-domain models consistently outperformed the tested medical-domain models. While the medical models frequently generated hallucinated outputs and exhibited difficulties following task instructions, the general-domain models demonstrated moderate alignment with human annotations. Among them, Gemma3:12b achieved the strongest overall agreement, obtaining the highest correlation with human scores (Pearson = 0.51, Spearman = 0.59). Item-level analysis revealed substantial variation in performance across OPTION12 categories, suggesting that different aspects of shared decision making pose different levels of difficulty for current LLMs.

Beyond quantitative evaluation, we identified several recurring categories of LLM errors, including failures in temporal reasoning, role attribution, interpretation of decision-making options, evidence selection, and hallucinated justifications. These findings provide insights into the limitations of current OS-sLLMs and highlight specific challenges that must be addressed before reliable deployment in clinical quality assessment workflows.

Overall, our results suggest that open-source smaller language models are not yet capable of replacing human annotators for OPTION12 coding. However, they show potential

as supportive tools within human-in-the-loop annotation pipelines, particularly for accelerating coding and assisting quality assurance while preserving privacy through local deployment.

Several directions remain for future investigation. First, the optimized prompting strategies and model configurations identified during the development phase will be deployed on the held-out test set to evaluate the generalizability of the findings. This evaluation will also assess the effectiveness of the proposed Judge-LLM framework for resolving disagreements among multiple OS-sLLMs. Second, future work will investigate ensemble and consensus-based approaches that combine the complementary strengths of different models across OPTION12 items. The observed variability in item-level performance suggests that model collaboration may provide more reliable assessments than any individual model alone. Third, additional work is needed to improve medical-domain OS-sLLMs. Despite their domain specialization, the tested medical models exhibited substantial hallucination and instruction-following issues. Future experiments will examine whether prompt optimization, supervised fine-tuning, parameter-efficient adaptation methods (e.g., LoRA), or domain-specific instruction tuning can improve their reliability for SDM assessment tasks. Fourth, we plan to quantify inter-rater reliability between human annotators and compare LLM performance against human-human agreement levels. Such analysis will provide a more meaningful benchmark for evaluating the practical usefulness of LLM-assisted coding. Finally, future research will explore human-in-the-loop workflows in which LLMs provide preliminary OPTION12 annotations and evidence extraction while human experts verify and refine the outputs. This approach may reduce annotation burden while maintaining the transparency and reliability required in clinical research settings.

Table 5: Qualitative analysis of LLM performance in shared decision making.

Issue Type	Item	LLM Interpretation	Human Annotation	Consensus
Temporality-management	Item 1: The clinician draws attention to an identified problem as one that requires a decision making process	LLM identifies the decision-making process at the end of the consultation rather than at the beginning. For example, Gemma and Mistral use the clinician's statement about discussing treatment as evidence, even though the decision framing occurs too late in the consultation.	Human-1: 1; 0 Human-2: 2	
Misinterpreting the item/option	Item 2: The clinician states that there is more than one way to deal with the identified problem	Gemma cites explanation of melanoma and Breslow thickness as if it were discussion of multiple management options, but the quoted evidence does not present alternatives.	Human-1: 1; 1 Human-2: 1	
Misinterpreting the item/option (continued)	Item 1	Llama 3.1 treats the clinician's check of what was previously explained about melanoma as if it were introducing a decision process, while it is actually background clarification.	Human-1: 1; 0 Human-2: 2	
Scoring right but justification/evidence not accurate	Item 3: The clinician assesses the patient's preferred approach to receiving information to assist decision making	Mistral assigns a low score appropriately, but the justification and evidence focus on explanation of Breslow thickness rather than the patient's preferred way of receiving information.	Human-1: 1; 1 Human-2: 1	
Scoring right but evidence not accurate	Item 4: Clinician lists all options, including doing nothing	Mistral's justification correctly notes that treatment options were not explicitly discussed, but the selected evidence sentence is unrelated and appears arbitrary.	Human annotation: no listing; not detailed about wait-and-see	1
Justification / evidence right but score differs from human	Item 1 (File 130003)	Gemma, Llama, and Mistral correctly identify that the clinician introduces the treatment discussion, but they over-score it as full decision framing or shared decision making.	Human: 1, 1, 1	1

Continued on next page

Issue Type	Item	LLM Interpretation	Human Annotation	Consensus
Missing justification/evidence	Item 3 (File 130003)	Gemma assigns score 1 and states that there is no explicit assessment of the patient's preferred learning style, but provides no actual evidence excerpt.	Human: 1, 1, 1	1
Role misunderstanding	Item 8: The clinician checks that the patient has understood the information	Mistral uses a caregiver utterance as if it were spoken by the clinician, leading to an incorrect attribution of understanding-check behavior.	Human: 0, 0, 0	0
Hallucinated evidence	General issue	In some cases, the LLM produces evidence or statements that are not grounded in the transcript at all.	–	–
Evidence / justification partly right but score too high	Item 1 / later example from File 130003	The clinician asks for agreement with surgery, and the evidence is relevant, but the model over-interprets this as balanced shared decision making, whereas the interaction is more directive.	Human: 2, 2, 2	2
Evidence / justification okay but score too high	Continued example	Llama scores the clinician's brief acknowledgment of patient anxiety too highly, although the concern is only superficially explored.	Human: 1, 1, 1	1

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A Data Statistics vs SDM

We are curious about whether the speaker frequency, duration, and ratio from (doctor, patient, caregiver) matters regarding shared decision making outcomes. To investigate this aspect, we display the statistics of such.

Table 6 presents speaker-level descriptive statistics, including speaking turns, speaking duration, duration ratio, text length, total consultation duration, and total number of segments for the seven development-set consultations.

Looking into the last column on the right-hand side of Human SDM Mean, as well as the statistics from this table, the higher SDM scores are achieved by patient files 110022 and 130004 (SDM both 1.42), and 130005 (SDM 1.17). They have the doctor speaking ratio (70.5, 80.8, 62.5), which is a wide range. The other files with lower SDM scores have doctor speaking ratio values falling in the range [62.5, 80.8], so we can not make a conclusion on the impact of speaker turns on SDM scores.

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File	Speaker	Turns	Duration	Ratio (%)	Text Len.	Total Dur.	Segments	Human SDM Mean
110019	Doctor	104	13m 38s	49.9	14156	27m 18s	272	0.83
110019	Patient	97	7m 01s	25.7	3777	27m 18s	272	0.83
110019	Caregiver	71	6m 39s	24.4	5168	27m 18s	272	0.83
110022	Doctor	70	11m 02s	70.5	8091	15m 39s	155	1.42
110022	Patient	61	3m 46s	24.1	2086	15m 39s	155	1.42
110022	Caregiver	24	0m 51s	5.4	405	15m 39s	155	1.42
130002	Doctor	87	20m 23s	70.6	19461	28m 52s	204	0.67
130002	Patient	41	3m 10s	11.0	2053	28m 52s	204	0.67
130002	Caregiver	76	5m 19s	18.4	3646	28m 52s	204	0.67
130003	Doctor	79	14m 09s	67.5	13757	20m 58s	176	1.00
130003	Patient	67	4m 59s	23.8	4054	20m 58s	176	1.00
130003	Caregiver	30	1m 50s	8.7	1390	20m 58s	176	1.00
130004	Doctor	35	8m 40s	80.8	7759	10m 44s	73	1.42
130004	Patient	20	0m 50s	7.8	427	10m 44s	73	1.42
130004	Caregiver	18	1m 14s	11.5	502	10m 44s	73	1.42
130005	Doctor	86	10m 17s	62.5	11496	16m 28s	173	1.17
130005	Patient	87	6m 11s	37.5	4147	16m 28s	173	1.17
130006	Doctor	45	9m 13s	69.1	6973	13m 20s	86	0.42
130006	Patient	41	4m 07s	30.9	1202	13m 20s	86	0.42

Table 6: Speaker-level descriptive statistics with corresponding human-annotated SDM mean scores (OPTION12) for each consultation.

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